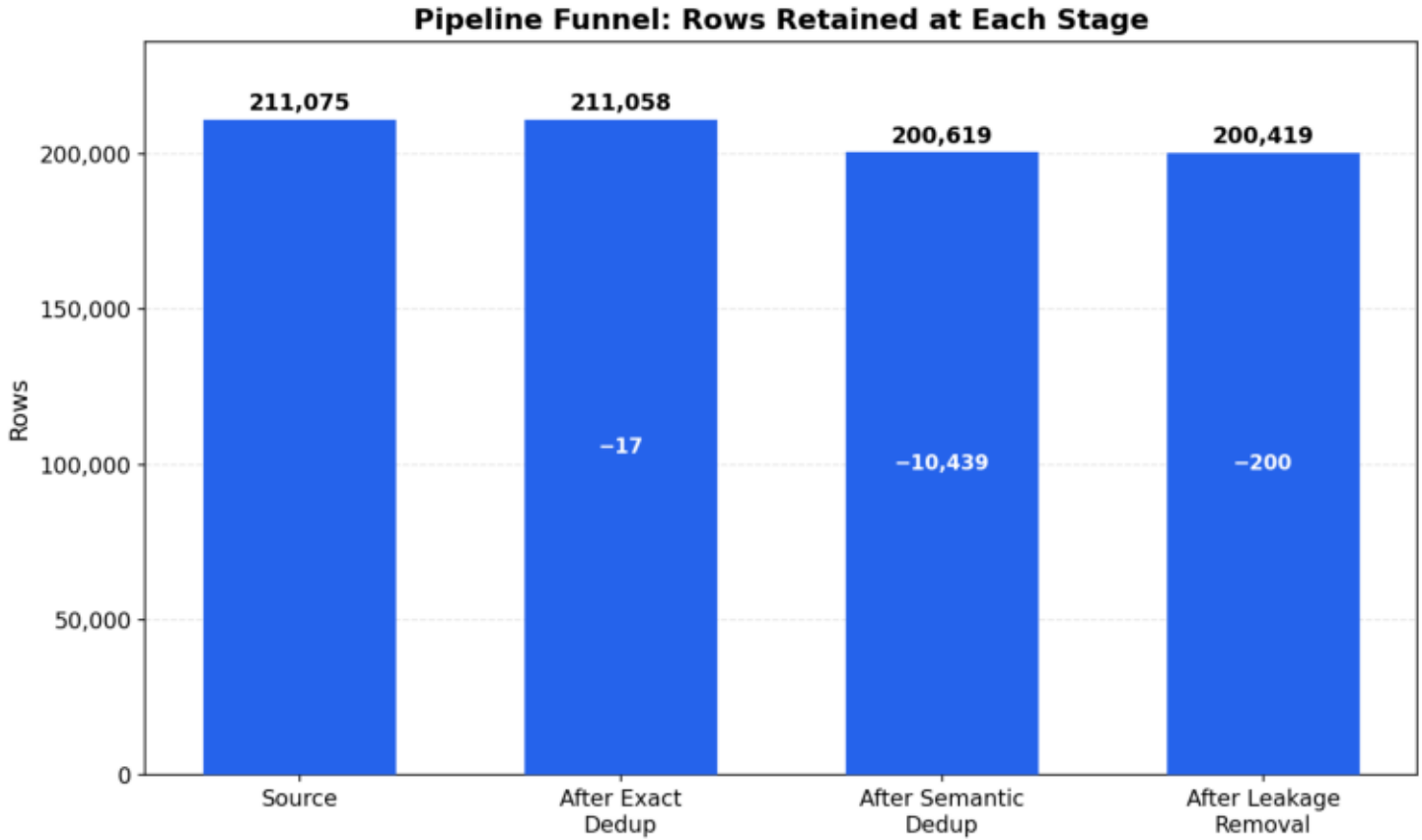
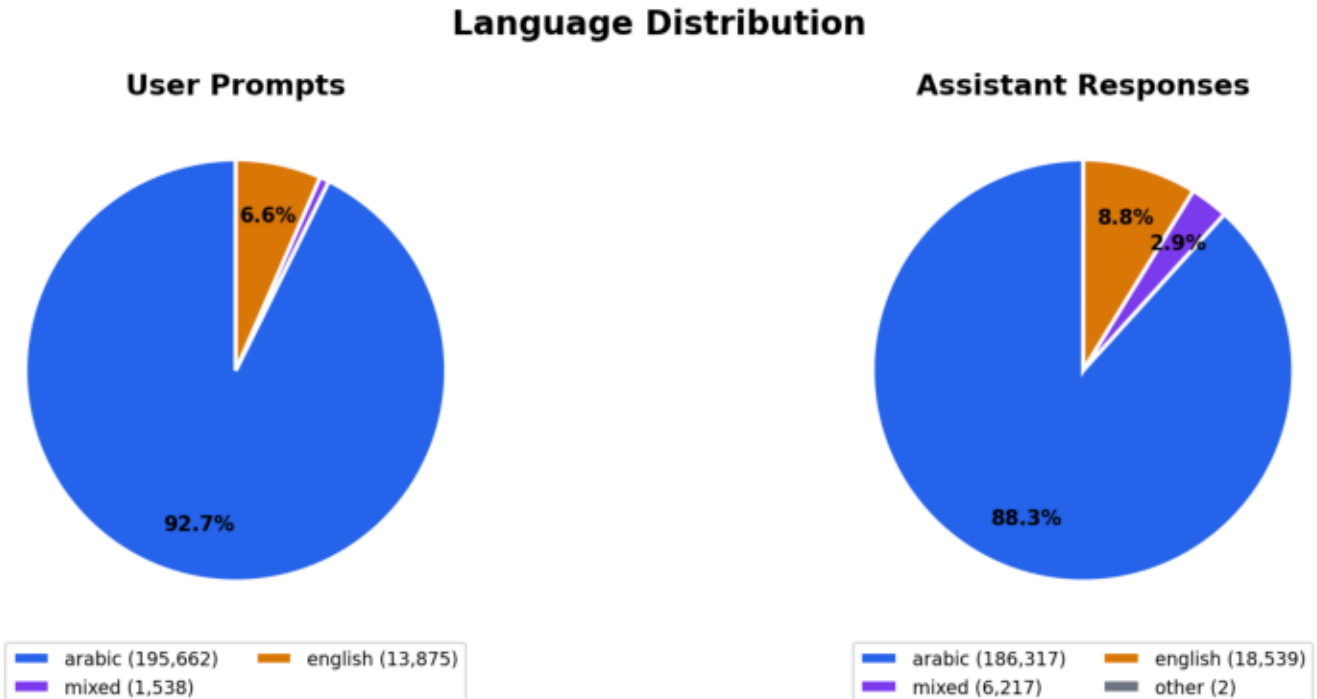
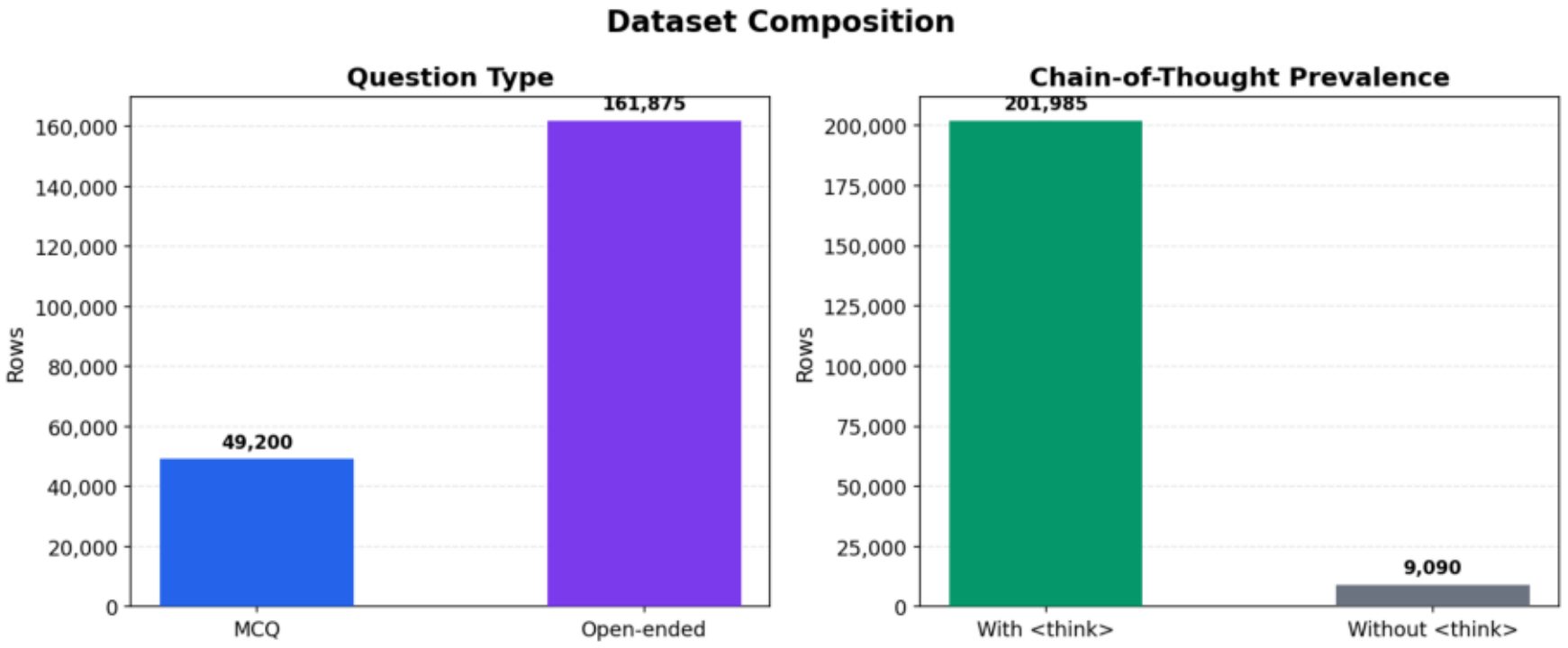
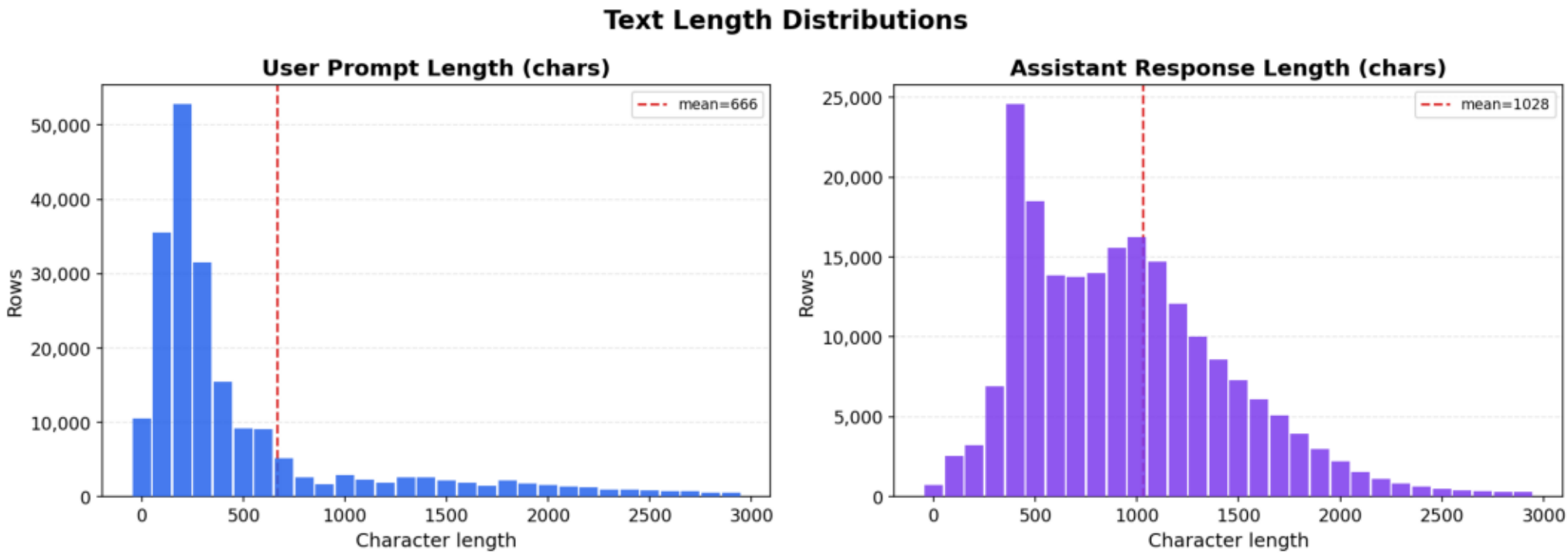
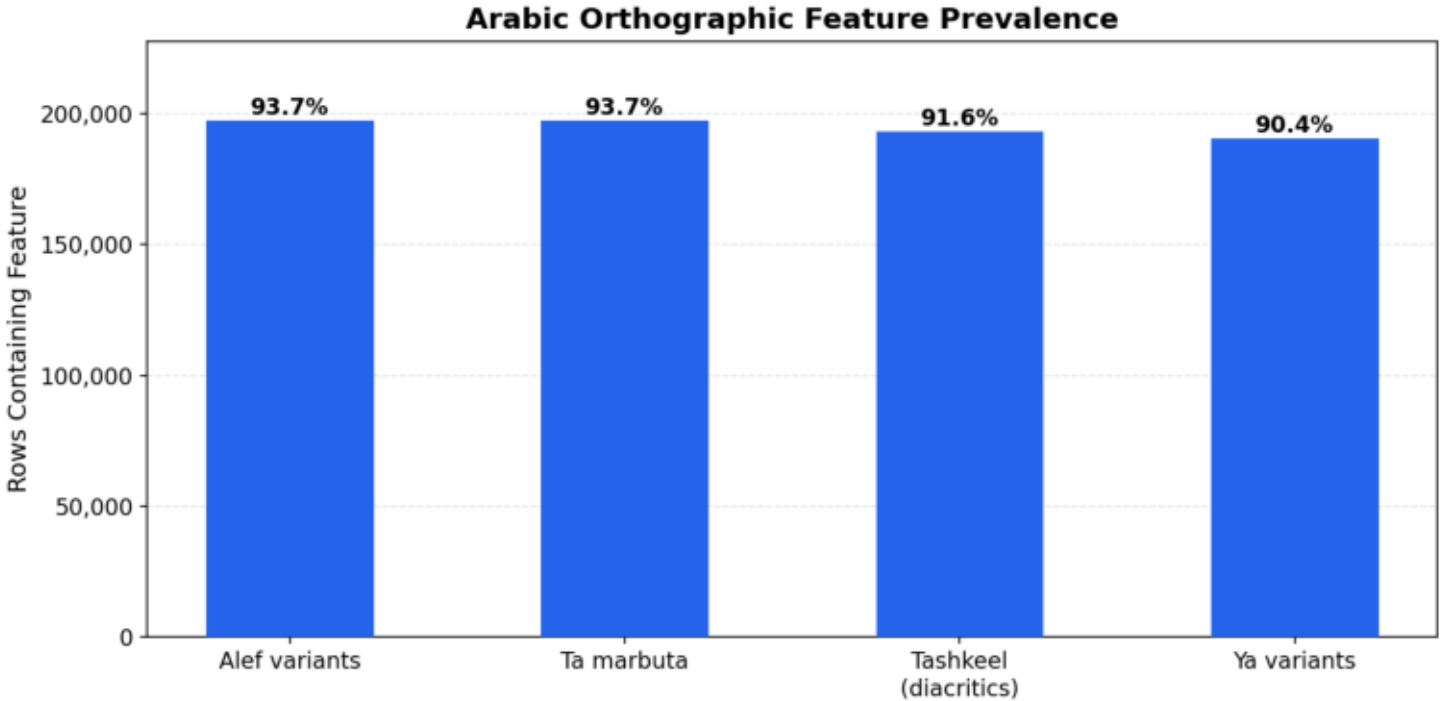
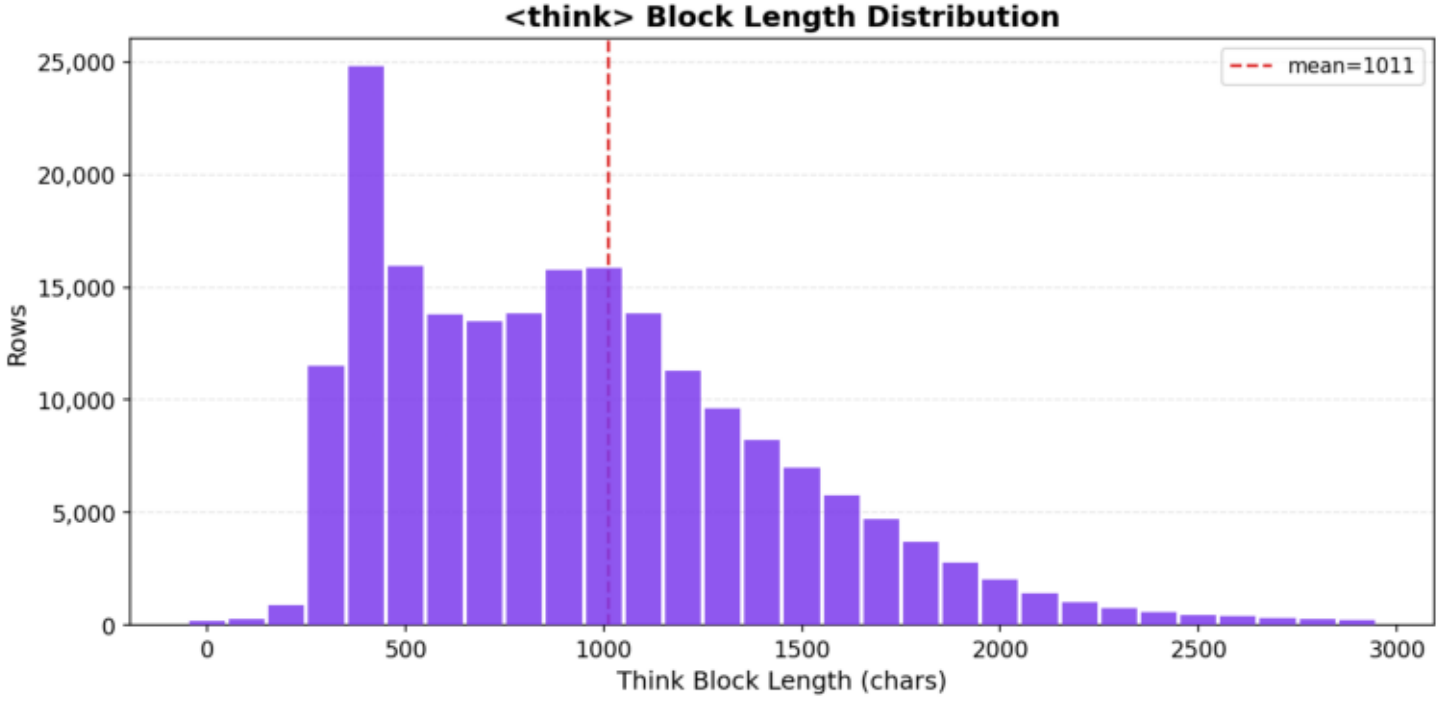

Pipeline Funnel: Rows Retained at Each Stage
Rows
0
50,000
100,000
150,000
200,000
211,075
211,058
200,619
200,419
−17
−10,439
−200
Source
After Exact Dedup
After Semantic Dedup
After Leakage Removal

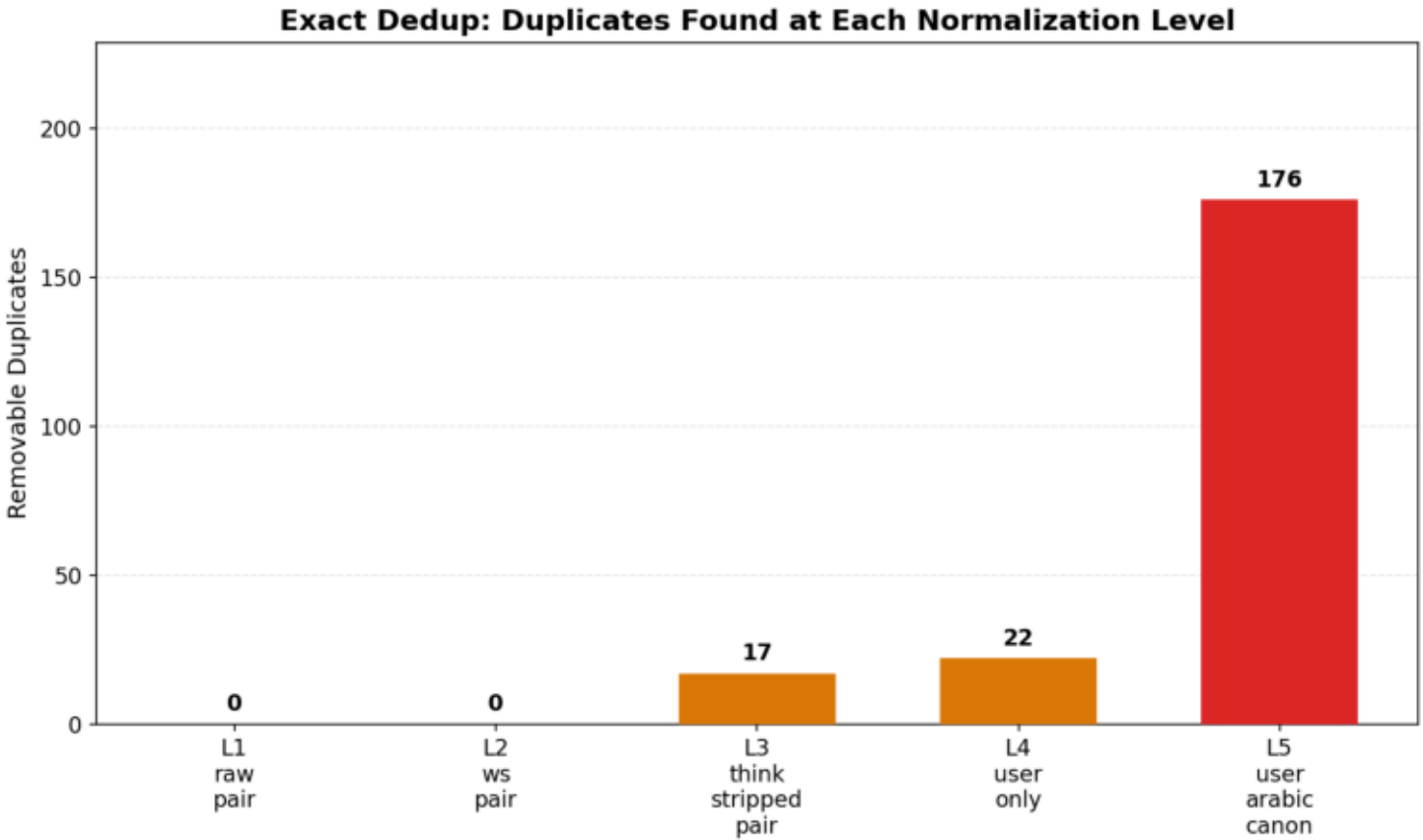

Language Distribution
User Prompts
6.6%
92.7%
arabic (195,662)
english (13,875)
mixed (1,538)
Assistant Responses
8.8%
2.9%
88.3%
arabic (186,317)
english (18,539)
mixed (6,217)
other (2)

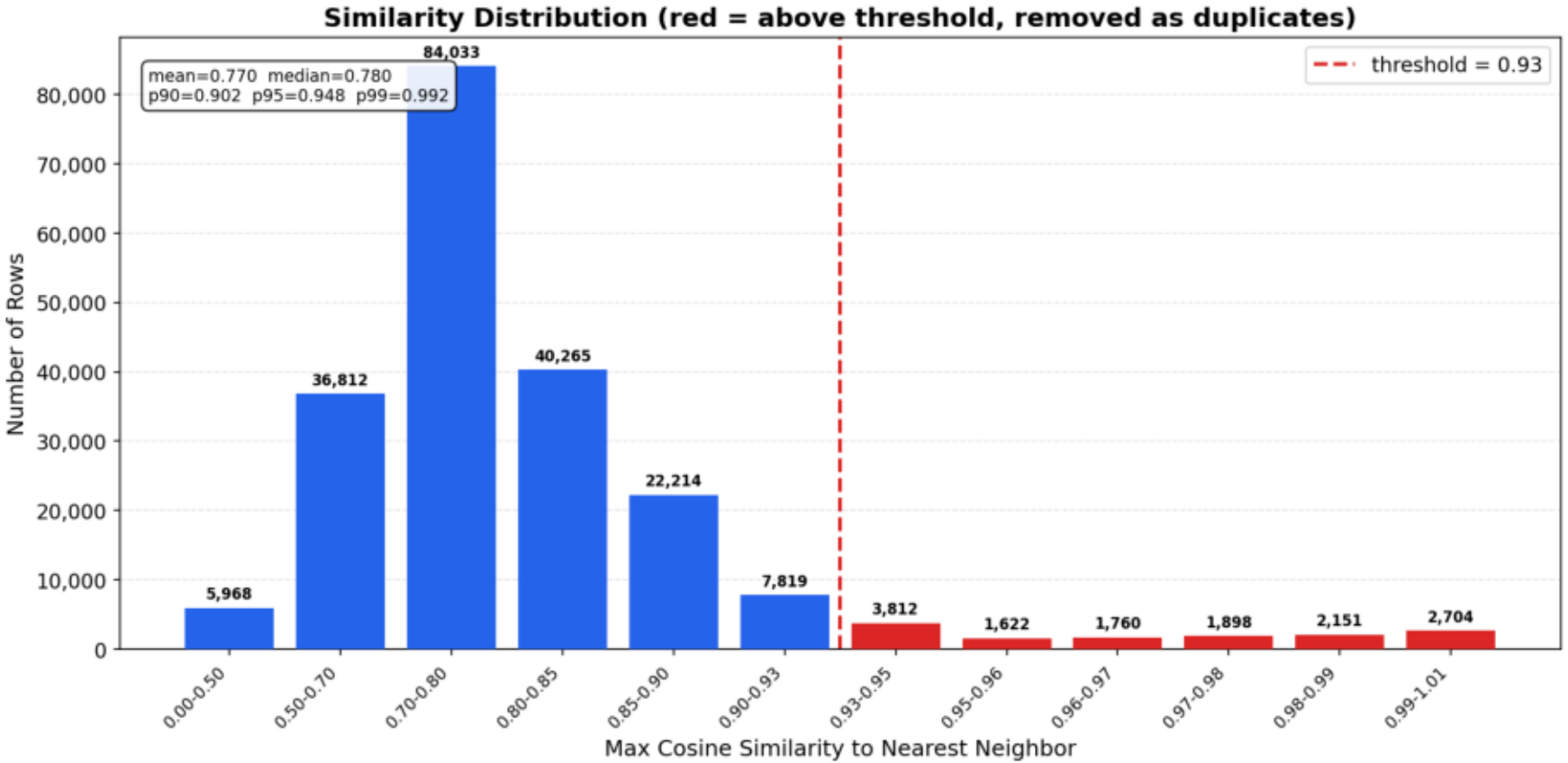

Dataset Composition
Question Type
Rows
160,000
140,000
120,000
100,000
80,000
60,000
40,000
20,000
0
49,200
161,875
MCQ
Open-ended
Chain-of-Thought Prevalence
Rows
200,000
175,000
150,000
125,000
100,000
75,000
50,000
25,000
0
201,985
9,090
With <think>
Without <think>

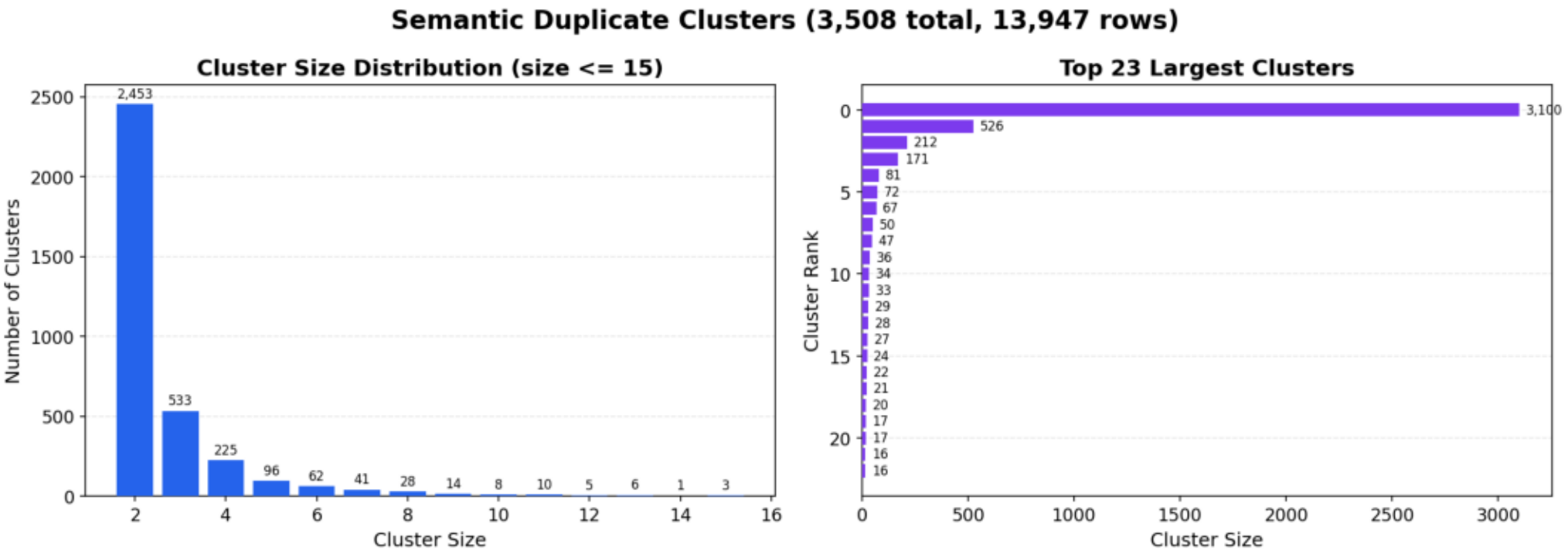

Text Length Distributions
User Prompt Length (chars)
mean=666
Rows
Character length
0
10,000
20,000
30,000
40,000
50,000
0
500
1000
1500
2000
2500
3000
Assistant Response Length (chars)
mean=1028
Rows
Character length
0
5,000
10,000
15,000
20,000
25,000
0
500
1000
1500
2000
2500
3000

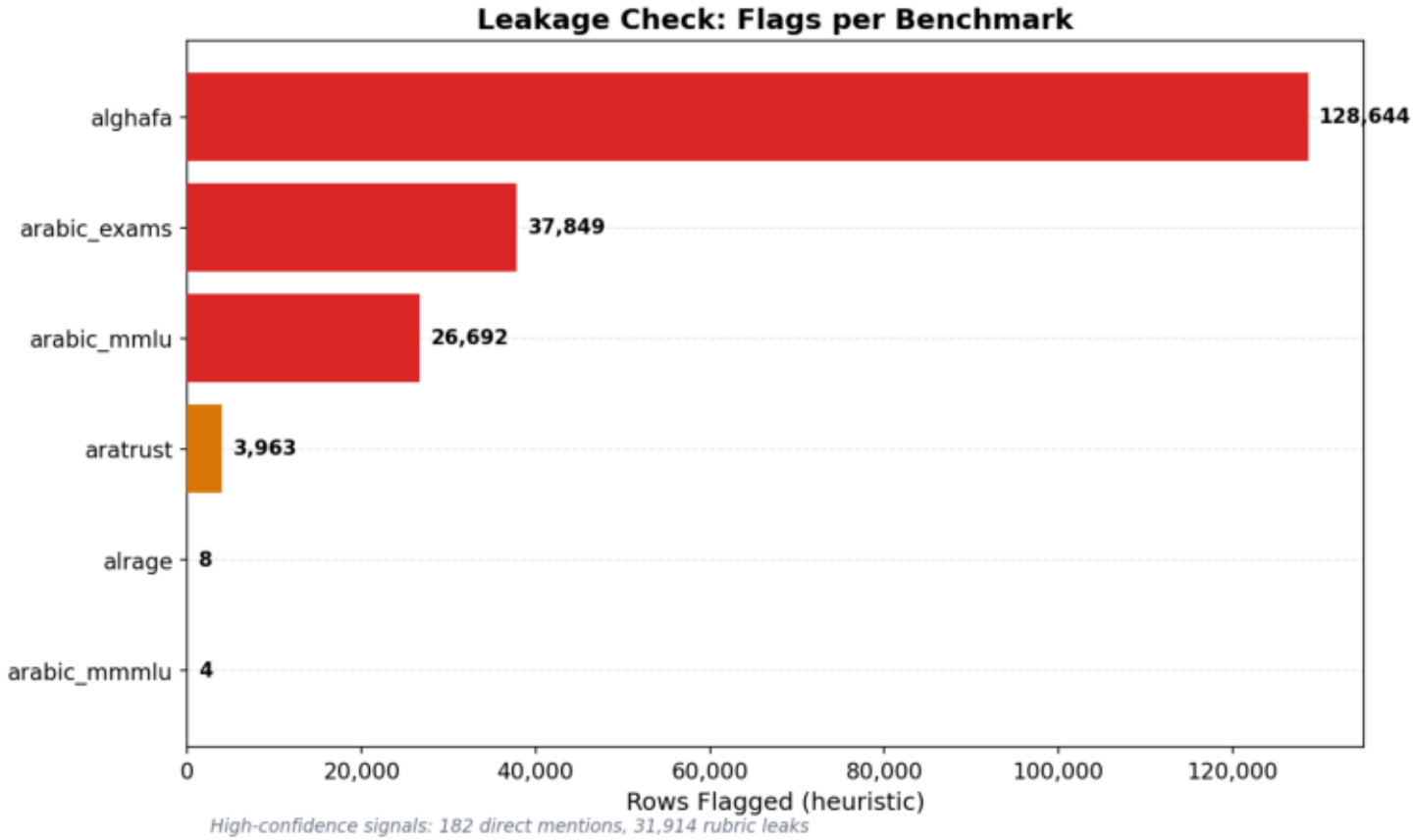

Arabic Orthographic Feature Prevalence
Rows Containing Feature
200,000
150,000
100,000
50,000
0
93.7%
93.7%
91.6%
90.4%
Alef variants
Ta marbuta
Tashkeel
(diacritics)
Ya variants

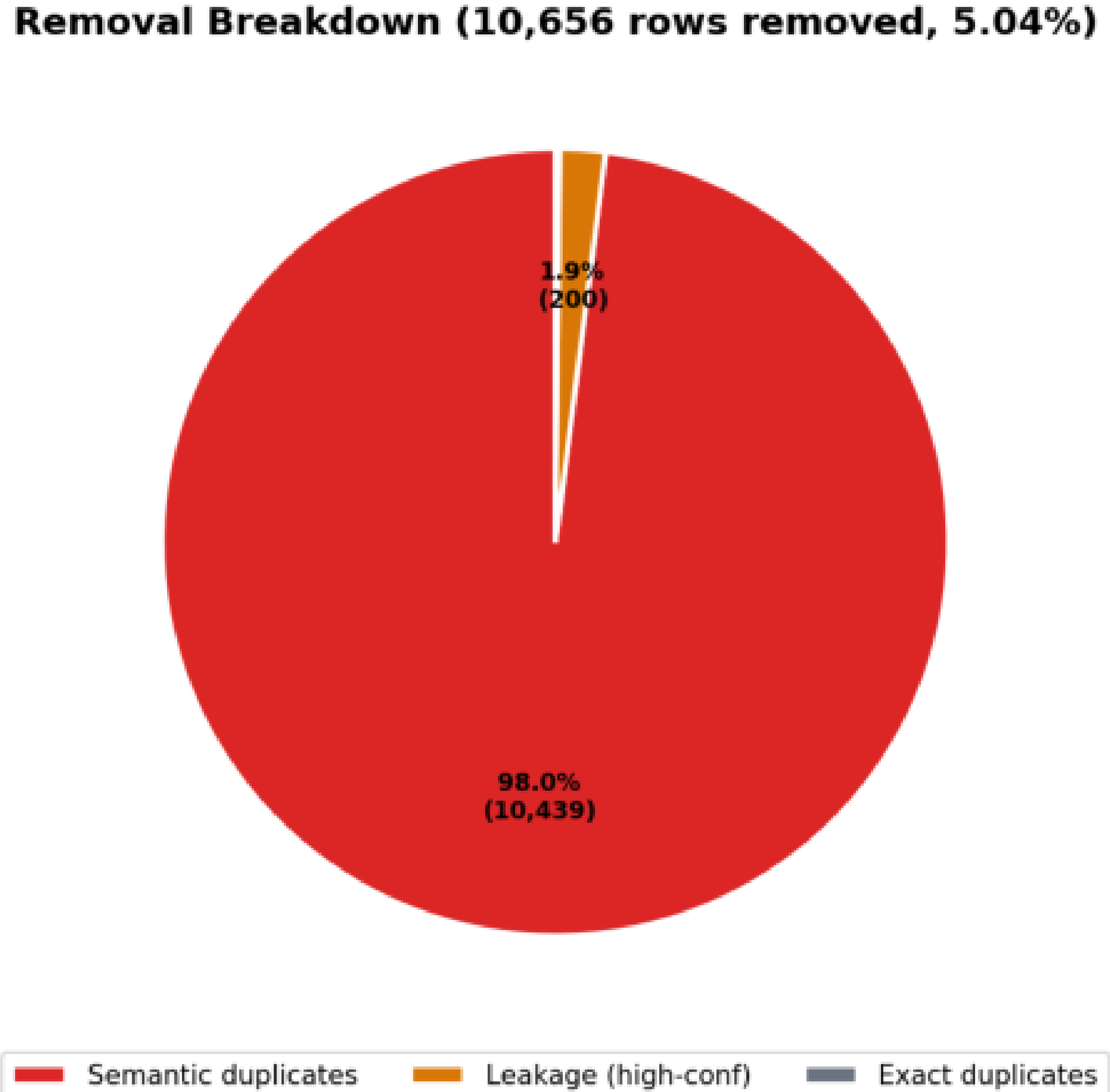

<think> Block Length Distribution
mean=1011
Rows
25,000
20,000
15,000
10,000
5,000
0
0
500
1000
1500
2000
2500
3000
Think Block Length (chars)


Exact Dedup: Duplicates Found at Each Normalization Level
Removable Duplicates
0
50
100
150
200
0
0
17
22
176
L1 raw pair
L2 ws pair
L3 think stripped pair
L4 user only
L5 user arabic canon


Similarity Distribution (red = above threshold, removed as duplicates)
mean=0.770 median=0.780
p90=0.902 p95=0.948 p99=0.992
threshold = 0.93
Number of Rows
0
10,000
20,000
30,000
40,000
50,000
60,000
70,000
80,000
5,968
36,812
84,033
40,265
22,214
7,819
3,812
1,622
1,760
1,898
2,151
2,704
0.00-0.50
0.50-0.70
0.70-0.80
0.80-0.85
0.85-0.90
0.90-0.93
0.93-0.95
0.95-0.96
0.96-0.97
0.97-0.98
0.98-0.99
0.99-1.01
Max Cosine Similarity to Nearest Neighbor


Semantic Duplicate Clusters (3,508 total, 13,947 rows)
Cluster Size Distribution (size <= 15)
Number of Clusters
Cluster Size
2,453
533
225
96
62
41
28
14
8
10
5
6
1
3
Top 23 Largest Clusters
Cluster Rank
Cluster Size
3,100
526
212
171
81
72
67
50
47
36
34
33
29
28
27
24
22
21
20
17
17
16
16


Leakage Check: Flags per Benchmark
alghafa
arabic_exams
arabic_mmlu
aratrust
alrage
arabic_mmmlu
128,644
37,849
26,692
3,963
8
4
0
20,000
40,000
60,000
80,000
100,000
120,000
Rows Flagged (heuristic)
High-confidence signals: 182 direct mentions, 31,914 rubric leaks


Removal Breakdown (10,656 rows removed, 5.04%)
1.9%
(200)
98.0%
(10,439)
Semantic duplicates
Leakage (high-conf)
Exact duplicates